Özyeğin University

Math 217 - Probability and Statistics

In Class Exercise - 3

.

Question 1: In the New Jersey state lottery, each $1 ticket has six randomly marked numbers out of 1,…,46. A ticket is a winner if the six marked numbers match six numbers drawn at random at the end of a week. For each ticket sold, 50 cents is added to the pot for the winners. If there are k winning tickets, the pot is divided equally among the winners. Suppose you bought a winning ticket in a week in which 2n tickets are sold, and the pot is n dollars.

a) What is the probability q that a random ticket will be winner?
b) Find the PMF of Kn, the number of other (besides your own) winning tickets.
c) What is the expected value of Wn, the prize for winning ticket?

If there is no winner for the lottery, then the pot is carried over the next week. Suppose that in a given week, an r dollar pot is carried over from the previous week and 2n tickets sold.

d) If you own one of the 2n tickets sold, what is the expected value of V, the value (i.e., the amount you win) of that ticket? Is it ever possible that E(V)>1?
e) Suppose that before the ticket sales are stopped, you are given the opportunity to buy one of each possible ticket. For what values (if any) of n and r should you do it?

Question 2: In real-time packet data transmission, the time between successfully received packets is called the interarrival time, and randomness in packet interarrival times is called jitter. Jitter is undesriable. One measure of jitter is the standard deviation of the packet interarrival time. From the following problem asked in class-exercise 2, calculate the jitter variance (Var(T)) . How large must the successful transmission probability q be to ensure that the jitter is less than 2 milliseconds

"A source transmits data packets to a receiver over a network link. The receiver uses detection to identify packets that have been corrupted by noise. When a packet is received error-free, the receiver sends an acknowledgment (ACK) back to the source. When the receiver gets a packet with errors, a negative acknowledgment (NAK) message is sent back to the source. Each time the source receives a NAK, the packet is retransmitted. We assume that each packet transmission is independently corrupted by errors with probability q.

a) Find the PMF of X, the number of times that a packet is trasnmitted by the source.

b) Suppose each packet takes one millisecond to transmit and that the source waits an additional millisecond to receive the acknowledgment message (ACK or NAK) before retransmitting. Let T equal the time required until the packet is successfully received. What is the relationship between T and X? What is the PMF of T?"

Sol:

1)

(a) There are $\binom{46}{6}$ equally likely winning combinations so that

$$q = \frac{1}{\binom{46}{6}} = \frac{1}{9{,}366{,}819} \approx 1.07 \times 10^{-7}. \tag{1}$$

(b) Assuming each ticket is chosen randomly, each of the $2n-1$ other tickets is independently a winner with probability q. The number of other winning tickets K_n has the binomial PMF

$$P_{K_n}(k) = \begin{cases} \binom{2n-1}{k} q^k (1-q)^{2n-1-k} & k = 0, 1, \ldots, 2n-1, \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

(c) Since there are $K_n + 1$ winning tickets in all, the value of your winning ticket is $W_n = n/(K_n+1)$ which has mean

$$\mathrm{E}[W_n] = n\,\mathrm{E}\left[\frac{1}{K_n+1}\right]. \tag{3}$$

Calculating the expected value

$$\mathrm{E}\left[\frac{1}{K_n+1}\right] = \sum_{k=0}^{2n-1} \left(\frac{1}{k+1}\right) P_{K_n}(k) \tag{4}$$

is fairly complicated. The trick is to express the sum in terms of the sum of a binomial PMF.

$$\begin{aligned} \mathrm{E}\left[\frac{1}{K_n+1}\right] &= \sum_{k=0}^{2n-1} \frac{1}{k+1} \frac{(2n-1)!}{k!(2n-1-k)!} q^k (1-q)^{2n-1-k} \\ &= \frac{1}{2n} \sum_{k=0}^{2n-1} \frac{(2n)!}{(k+1)!(2n-(k+1))!} q^k (1-q)^{2n-(k+1)}. \end{aligned} \tag{5}$$

By factoring out $1/q$, we obtain

$$\begin{aligned} \mathrm{E}\left[\frac{1}{K_n+1}\right] &= \frac{1}{2nq} \sum_{k=0}^{2n-1} \binom{2n}{k+1} q^{k+1} (1-q)^{2n-(k+1)} \\ &= \frac{1}{2nq} \underbrace{\sum_{j=1}^{2n} \binom{2n}{j} q^j (1-q)^{2n-j}}_{A}. \end{aligned} \tag{6}$$

We observe that the above sum labeled A is the sum of a binomial PMF for $2n$ trials and success probability q over all possible values except $j = 0$. Thus

$$A = 1 - \binom{2n}{0} q^0 (1-q)^{2n-0} = 1 - (1-q)^{2n}. \tag{7}$$

This implies

$$\mathrm{E}\left[\frac{1}{K_n + 1}\right] = \frac{1 - (1-q)^{2n}}{2nq}. \tag{8}$$

Our expected return on a winning ticket is

$$\mathrm{E}[W_n] = n\,\mathrm{E}\left[\frac{1}{K_n + 1}\right] = \frac{1 - (1-q)^{2n}}{2q}. \tag{9}$$

Note that when $nq \ll 1$, we can use the approximation that $(1-q)^{2n} \approx 1 - 2nq$ to show that

$$\mathrm{E}[W_n] \approx \frac{1 - (1 - 2nq)}{2q} = n, \qquad nq \ll 1. \tag{10}$$

However, in the limit as the value of the prize n approaches infinity, we have

$$\lim_{n\to\infty} \mathrm{E}[W_n] = \frac{1}{2q} \approx 4.683 \times 10^6. \tag{11}$$

That is, as the pot grows to infinity, the expected return on a winning ticket doesn't approach infinity because there is a corresponding increase in the number of other winning tickets. If it's not clear how large n must be for this effect to be seen, consider the following table:

n	10^6	10^7	10^8
$\mathrm{E}[W_n]$	9.00×10^5	4.13×10^6	4.68×10^6

(12)

When the pot is \$1 million, our expected return is \$900,000. However, we see that when the pot reaches \$100 million, our expected return is very close to $1/(2q)$, less than \$5 million!

for d)-f)

(a) There are $\binom{46}{6}$ equally likely winning combinations so that

$$q = \frac{1}{\binom{46}{6}} = \frac{1}{9{,}366{,}819} \approx 1.07 \times 10^{-7}. \tag{1}$$

(b) Assuming each ticket is chosen randomly, each of the $2n-1$ other tickets is independently a winner with probability q. The number of other winning tickets K_n has the binomial PMF

$$P_{K_n}(k) = \begin{cases} \binom{2n-1}{k} q^k (1-q)^{2n-1-k} & k = 0, 1, \ldots, 2n-1, \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

Since the pot has $n+r$ dollars, the expected amount that you win on your ticket is

$$\mathrm{E}[V] = 0(1-q) + q\,\mathrm{E}\left[\frac{n+r}{K_n+1}\right] = q(n+r)\,\mathrm{E}\left[\frac{1}{K_n+1}\right]. \tag{3}$$

Note that $\mathrm{E}[1/K_n+1]$ was also evaluated in Problem 3.7.10. For completeness, we repeat those steps here.

$$\begin{aligned} \mathrm{E}\left[\frac{1}{K_n+1}\right] &= \sum_{k=0}^{2n-1} \frac{1}{k+1} \frac{(2n-1)!}{k!(2n-1-k)!} q^k (1-q)^{2n-1-k} \\ &= \frac{1}{2n} \sum_{k=0}^{2n-1} \frac{(2n)!}{(k+1)!(2n-(k+1))!} q^k (1-q)^{2n-(k+1)}. \end{aligned} \tag{4}$$

By factoring out $1/q$, we obtain

$$\begin{aligned} \mathrm{E}\left[\frac{1}{K_n+1}\right] &= \frac{1}{2nq} \sum_{k=0}^{2n-1} \binom{2n}{k+1} q^{k+1} (1-q)^{2n-(k+1)} \\ &= \frac{1}{2nq} \underbrace{\sum_{j=1}^{2n} \binom{2n}{j} q^j (1-q)^{2n-j}}. \end{aligned} \tag{5}$$

We observe that the above sum labeled A is the sum of a binomial PMF for $2n$ trials and success probability q over all possible values except $j = 0$. Thus $A = 1 - \binom{2n}{0} q^0 (1-q)^{2n-0}$, which implies

$$\mathrm{E}\left[\frac{1}{K_n + 1}\right] = \frac{A}{2nq} = \frac{1 - (1-q)^{2n}}{2nq}. \tag{6}$$

The expected value of your ticket is

$$\begin{aligned} \mathrm{E}[V] &= \frac{q(n+r)[1-(1-q)^{2n}]}{2nq} \\ &= \frac{1}{2}\left(1 + \frac{r}{n}\right)[1-(1-q)^{2n}]. \end{aligned} \tag{7}$$

Each ticket tends to be more valuable when the carryover pot r is large and the number of new tickets sold, $2n$, is small. For any fixed number n, corresponding to $2n$ tickets sold, a sufficiently large pot r will guarantee that $\mathrm{E}[V] > 1$. For example if $n = 10^7$, (20 million tickets sold) then

$$\mathrm{E}[V] = 0.44\left(1 + \frac{r}{10^7}\right). \tag{8}$$

If the carryover pot r is 30 million dollars, then $\mathrm{E}[V] = 1.76$. This suggests that buying a one dollar ticket is a good idea. This is an unusual situation because normally a carryover pot of 30 million dollars will result in far more than 20 million tickets being sold.

So that we can use the results of the previous part, suppose there were $2n - 1$ tickets sold before you must make your decision. If you buy one of each possible ticket, you are guaranteed to have one winning ticket. From the other $2n - 1$ tickets, there will be K_n winners. The total number of winning tickets will be $K_n + 1$. In the previous part we found that

$$\mathrm{E}\left[\frac{1}{K_n + 1}\right] = \frac{1 - (1-q)^{2n}}{2nq}. \tag{9}$$

Let R denote the expected return from buying one of each possible ticket. The pot had r dollars beforehand. The $2n-1$ other tickets are sold add $n-1/2$ dollars to the pot. Furthermore, you must buy $1/q$ tickets, adding $1/(2q)$ dollars to the pot. Since the cost of the tickets is $1/q$ dollars, your expected profit

$$\begin{aligned}
\mathrm{E}[R] &= \mathrm{E}\left[\frac{r+n-1/2+1/(2q)}{K_n+1}\right] - \frac{1}{q} \\
&= \frac{q(2r+2n-1)+1}{2q}\,\mathrm{E}\left[\frac{1}{K_n+1}\right] - \frac{1}{q} \\
&= \frac{[q(2r+2n-1)+1](1-(1-q)^{2n})}{4nq^2} - \frac{1}{q}. \qquad (10)
\end{aligned}$$

For fixed n, sufficiently large r will make $\mathrm{E}[R] > 0$. On the other hand, for fixed r, $\lim_{n\to\infty} \mathrm{E}[R] = -1/(2q)$. That is, as n approaches infinity, your expected loss will be quite large.

2)

With our measure of jitter being σ_T, and the fact that $T = 2X - 1$, we can express the jitter as a function of q by realizing that

$$\mathrm{Var}[T] = 4\,\mathrm{Var}[X] = \frac{4q}{(1-q)^2}. \qquad (1)$$

Therefore, our maximum permitted jitter is

$$\sigma_T = \frac{2\sqrt{q}}{(1-q)} = 2 \text{ ms}. \qquad (2)$$

Solving for q yields $q^2 - 3q + 1 = 0$. By solving this quadratic equation, we obtain

$$q = \frac{3 \pm \sqrt{5}}{2} = 3/2 \pm \sqrt{5}/2. \qquad (3)$$

Since q must be a value between 0 and 1, we know that a value of $q = 3/2 - \sqrt{5}/2 \approx 0.382$ will ensure a jitter of at most 2 milliseconds.